

**Precision is to numbers
what resolution is to
images**

```
graph TD; A["Precision is to numbers  
what resolution is to  
images"] --> B["float16  
half precision  
smallest safe gap: 1e-3"]; A --> C["float32  
single precision  
smallest safe gap: 1e-7"]; A --> D["float64  
double precision  
smallest safe gap: 1e-16"];
```

float16

half precision

smallest safe gap: $1e-3$

float32

single precision

smallest safe gap: $1e-7$

float64

double precision

smallest safe gap: $1e-16$